ASSIGNMENT-1 (Computer System Architecture)

Instructions:

1. *Submit the handwritten assignment by Monday (10-Feb-2025) without any delay.*
2. *Make sure to add your name and roll number on the top of every assignment page.*
3. *This is a graded assignment.*
4. *All questions are compulsory to attempt.*

Questions:

1. Implement a 2-bit adder using logic gates. Show the circuit diagram and truth table.
2. Convert the given Standard POS to Canonical POS:

   F(P,Q,R) = (P+Q).(Q+R).(P+R)
3. Minimize using Karnaugh Map:

   F(A,B,C,D) = Л(0,2,5,7,9,11)
4. Perform the following:
   a. 1010 + 1100
   b. $(11101.00110)_2$ = $(\ )_8$ = $(\ )_{10}$
5. Draw the block diagram of JK flip-flop along with its truth table, characteristic table and excitation table.
6. What are the different types of Registers?
7. What is the purpose of a program counter (PC) register in a computer system?
8. Describe the different types of buses in a computer system with the help of a diagram. Explain their functions and importance.
9. Give two differences between Combinational Circuit and Sequential Circuit.
10. Implement Full Adder using Half Adders.